In [5]:
```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.preprocessing import StandardScaler
from sklearn.cluster import KMeans,AgglomerativeClustering
from sklearn.metrics import silhouette_score
```

In [6]:
```
data=pd.read_csv("/home/backup/ml_datasets/Mall_Customers.csv")
print(data.head())
```

```
   CustomerID  Gender  Age  Annual Income (k$)  Spending Score (1-100)
0           1    Male   19                  15                      39
1           2    Male   21                  15                      81
2           3  Female   20                  16                       6
3           4  Female   23                  16                      77
4           5  Female   31                  17                      40
```

In [7]:
```
X=data[["Age","Annual Income (k$)"]]
print(X.head())
```

```
   Age  Annual Income (k$)
0   19                  15
1   21                  15
2   20                  16
3   23                  16
4   31                  17
```

In [8]:
```
scalar=StandardScaler()
X_scaled=scalar.fit_transform(X)
```

In [9]:
```
kmeans=KMeans(n_clusters=5,random_state=42,n_init=10)
kmeans_labels=kmeans.fit_predict(X_scaled)
kmeans_score=silhouette_score(X_scaled,kmeans_labels)
print("K-means silhouette score:",round(kmeans_score,3))
```

```
K-means silhouette score: 0.393
```

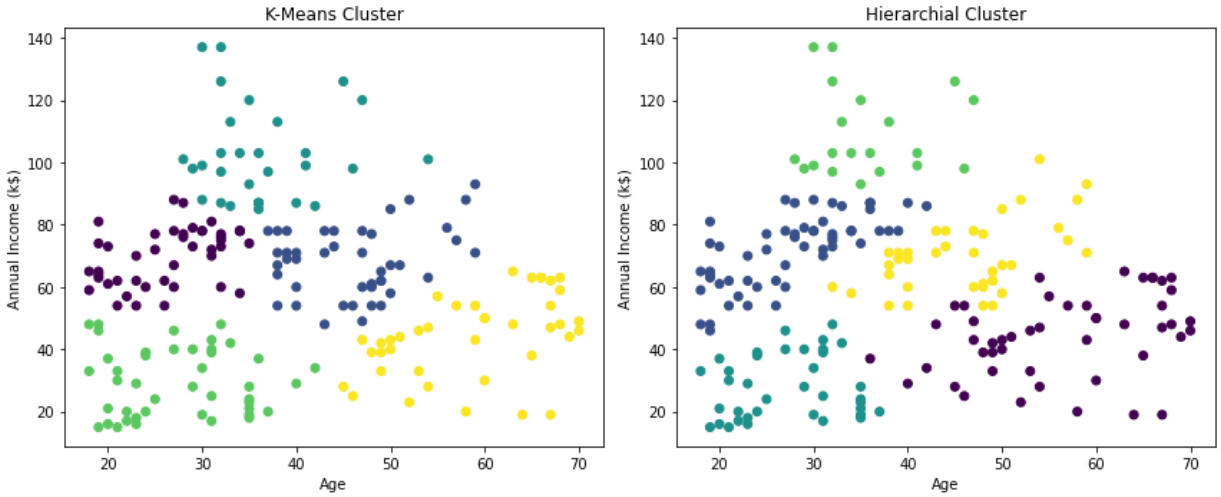
In [10]:
```
hierarchical=AgglomerativeClustering(n_clusters=5,linkage="ward")
hier_labels=hierarchical.fit_predict(X_scaled)
hier_score=silhouette_score(X_scaled,hier_labels)
print("Hierarchical silhouette score:",round(hier_score,3))
```

```
Hierarchical silhouette score: 0.358
```

In [11]:
```
plt.figure(figsize=(12,5))
plt.subplot(1,2,1)
plt.scatter(
    X['Age'],
    X['Annual Income (k$)'],
    c=kmeans_labels,
    cmap='viridis'
)
plt.title('K-Means Cluster')
plt.xlabel('Age')
plt.ylabel('Annual Income (k$)')

plt.subplot(1,2,2)
plt.scatter(
```

```
    X['Age'],
    X['Annual Income (k$)'],
    c=hier_labels,
    cmap='viridis'
)
plt.title('Hierarchial Cluster')
plt.xlabel('Age')
plt.ylabel('Annual Income (k$)')
plt.tight_layout()
plt.show()
```



In [12]:

```
print("/n Comparison")
print("K-Means silhouette Score:",round(kmeans_score,3))
print("Hierarchical Silhouette Score:",round(hier_score,3))
```

```
/n Comparison
K-Means silhouette Score: 0.393
Hierarchical Silhouette Score: 0.358
```

In [ ]: